SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 2.1: Introduction to Salts — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Introduction to Salts
Driving Question	DRIVING QUESTION: Why did the fertiliser granules get wet and clump together overnight, the washing soda crystals shrink and form a white crust, but the table salt stay completely the same — and what does understanding salts tell us about how to prepare and use them safely in Kenya?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **What Is a Salt? Classification and Types**	Students examine and compare three white solids: table salt (NaCl), calcium ammonium nitrate (CAN fertiliser), and washing soda ($Na_2CO_3 \cdot 10H_2O$). They record physical appearances and note that all three look broadly similar — white/off-white crystalline solids — yet behaved very differently overnight in the anchoring phenomenon. Teacher prompts: 'What do these three substances have in common? What differences did we already observe?' Students also examine provided solubility data and pH paper results for each substance dissolved in water.	Salts are ionic compounds formed when the hydrogen atom(s) in an acid are fully or partially replaced by a positive ion (a metal ion or ammonium ion). There are four types: (1) Normal salts — all replaceable hydrogen replaced, e.g., NaCl, $ZnSO_4$, CAN ($NH_4NO_3 \cdot Ca(NO_3)_2$); (2) Acid salts — hydrogen only partially replaced, e.g., $NaHCO_3$, $NaHSO_4$; (3) Basic salts — contain an OH^- group alongside the salt anion, e.g., Pb(OH)Cl; (4) Double salts — contain two different cations or anions, e.g., potash alum $KAl(SO_4)_2 \cdot 12H_2O$. Students learn to name salts from parent acids (hydrochloric → chloride; sulphuric → sulphate; nitric → nitrate; carbonic → carbonate; phosphoric → phosphate).	PEDAGOGICAL NOTE — This lesson builds the conceptual vocabulary for the entire sub-strand. Insist on precision: a salt is NOT simply 'what you get when acid meets base' — that is one method of preparation, not the definition. Use a two-column board organiser: 'Parent Acid | Salt Formed'. Cold-call at least 4 students to name the salt from a given acid + metal combination before moving on. Crucially, anchor back to the phenomenon: CAN fertiliser is a double salt (calcium nitrate + ammonium nitrate); NaCl is a normal salt; Na_2CO_3 is a normal salt of carbonic acid. Their different behaviours overnight are NOT yet explained — use this to populate the Driving Question Board. Wait time: pose 'Why might two normal salts behave so differently in air?' and give 30 seconds silent thinking before taking responses. Leave the question unresolved — it will be answered in Lesson 6.
Lesson 2 **Solubility of Salts and**	Students conduct a structured practical: small samples of named salts (NaCl,	Solubility determines which method can be used to prepare a salt — this is the critical	PEDAGOGICAL NOTE — Solubility rules are frequently examined and must be

the Solubility Rules	$PbSO_4$, $BaSO_4$, $CuCO_3$, AgCl, KNO_3, $Ca(OH)_2$) are added to water in test tubes and shaken. Students record whether each salt dissolves (soluble), partially dissolves (slightly soluble), or does not dissolve (insoluble). Results are tabulated on the board as a class data set. Teacher draws attention to patterns: all nitrates dissolved; most carbonates did not; sulphates were mixed.	decision-making tool for Lessons 3–5. Solubility rules for common salts: (1) All nitrates are soluble. (2) All sodium, potassium, and ammonium salts are soluble. (3) All chlorides are soluble EXCEPT AgCl and $PbCl_2$. (4) All sulphates are soluble EXCEPT $BaSO_4$, $PbSO_4$, and $CaSO_4$ (slightly soluble). (5) All carbonates are insoluble EXCEPT those of sodium, potassium, and ammonium. (6) All hydroxides are insoluble EXCEPT those of sodium, potassium, and ammonium ($Ca(OH)_2$ is slightly soluble). Students construct a solubility reference card they will use throughout the sub-strand.	memorised, but memorisation should follow pattern recognition from the practical. Do NOT give the rules before the experiment. After students tabulate results, ask: 'What pattern do you notice about nitrates? About carbonates?' Guide induction before providing formal rules. Use a mnemonic if one fits the class. Key teaching move: present an exam-style question — 'How would you prepare lead sulphate?' — and have students use their solubility card to realise $PbSO_4$ is insoluble, so the acid + metal method cannot work; this creates productive curiosity for Lesson 5. Anchor to phenomenon: 'Is CAN soluble? Is NaCl? Does solubility relate to what happened overnight?' (Partial answer: yes, solubility in air moisture is related but is not the same as solubility in bulk water — return to this in Lesson 6.)
Lesson 3 **Preparing Soluble Salts I — Acid + Metal and Acid + Metal Oxide (Experiment 2a)**	Students carry out Experiment 2a: preparation of zinc sulphate ($ZnSO_4$) crystals. Dilute H_2SO_4 is placed in a beaker; excess zinc granules are added. Students observe: effervescence (H_2 gas), gradual dissolution of zinc, the solution becoming clear (colourless). Excess zinc is filtered off. The filtrate is evaporated to reduce volume and allowed to crystallise. Students also observe a teacher demonstration of the acid + metal oxide route: CuO (black powder) added to warm dilute H_2SO_4; the black solid dissolves and the solution turns blue (Cu^{2+} ions). Students draw and annotate the experimental setups.	Two methods for preparing soluble salts: (1) Acid + Metal: $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$. Excess metal is used to ensure all acid reacts (confirmed when effervescence stops). Excess solid is removed by filtration. The salt is recovered by evaporation and crystallisation. Only metals above hydrogen in the reactivity series react; copper does not react with dilute acids this way. (2) Acid + Metal Oxide: $CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$. Excess metal oxide is used (no gas produced); excess solid removed by filtration; salt recovered by evaporation and crystallisation. Both methods use excess solid reagent to guarantee no acid remains in the final product.	PEDAGOGICAL NOTE — The concept of 'excess solid' is a key examination point. Pose the question: 'Why do we add excess zinc rather than excess acid?' Cold-call 3 students; expected answer: to ensure all acid is consumed so the final salt solution is pure. Emphasise the logic: the excess solid is easy to remove by filtration; excess dissolved acid is not. Step through the full procedure as a flow diagram on the board: React → Filter → Evaporate → Crystallise → Dry. Students should be able to reproduce this for any acid + metal or acid + metal oxide combination. Kenyan context: $ZnSO_4$ is used as a micronutrient fertiliser on Rift Valley tea farms; $CuSO_4$ is used in Bordeaux mixture (fungicide) on coffee and maize. Anchor to phenomenon: 'We now know how to make soluble salts like $ZnSO_4$ and $CuSO_4$. Is CAN made this way? Why or why not?' — sets up double salt discussion and links to applications in Lesson 7.
Lesson 4 **Preparing Soluble Salts II — Acid + Base and Acid +**	Students observe a teacher demonstration of titration-based salt preparation: NaOH solution (with phenolphthalein) is titrated against dilute HCl to find the exact volume	(1) Acid + Soluble Base (Neutralisation/Titration method): used when both the acid and the base are soluble (e.g., $HCl + NaOH \rightarrow NaCl + H_2O$).	PEDAGOGICAL NOTE — Students frequently confuse when to use titration vs. excess solid method. Use a decision flowchart: 'Is the base soluble? → Yes →

Carbonate/Hydrogen Carbonate	needed for neutralisation. The indicator-free repeat produces colourless NaCl(aq). Students then carry out a guided practical: Na_2CO_3 powder is added in small portions to dilute HCl in a beaker; they observe vigorous effervescence (CO_2), which subsides when carbonate is in excess. The mixture is filtered and evaporated to yield sodium chloride crystals. Students compare the two sub-methods and complete a comparison table.	An indicator is used in the first titration to find the exact endpoint; a second indicator-free titration gives a pure salt solution. No excess solid to filter — evaporation alone yields the salt. (2) Acid + Carbonate or Hydrogen Carbonate: $Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$; $NaHCO_3(s) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l) + CO_2(g)$. Excess carbonate (solid) is used to consume all acid; CO_2 effervescence ending signals completion; excess solid removed by filtration; salt recovered by evaporation. The critical selection rule: use titration ONLY when acid + base are both soluble; use excess solid method for all others.	Titration. No → Excess solid + filtration.' This flowchart, combined with the solubility rules card from Lesson 2, gives students a complete toolkit for Lessons 3–5. Common misconception to address explicitly: 'Can I just add a lot of NaOH and filter off the excess?' — No, because NaOH is soluble and cannot be filtered. Cold-call 4 students to classify given salt preparations (provide 4 examples mixing all methods seen so far). Kenyan context: washing soda (Na_2CO_3) — the substance in the anchoring phenomenon — can be traced to this preparation method; baking soda ($NaHCO_3$) familiar from mandazi preparation is an acid salt. Anchor to phenomenon: 'Table salt (NaCl) is made by the neutralisation method on an industrial scale. Does the method of preparation explain why it behaves differently in air?' — Probe: not directly, but purity of the product matters; leads toward Lesson 6.
Lesson 5 **Preparing Insoluble Salts — Double Decomposition and Precipitation**	Students carry out a practical: lead nitrate solution ($Pb(NO_3)_2$) and potassium iodide solution (KI) are mixed in a test tube. Students observe: immediate formation of a bright yellow precipitate (PbI_2). The mixture is filtered using a Büchner funnel or gravity filtration; the yellow solid is washed with distilled water and dried. Teacher demonstrates a second example: $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s)$ (white precipitate) + 2NaCl(aq). Students write ionic and net ionic equations for both reactions.	Insoluble salts are prepared by double decomposition (precipitation): two soluble salt solutions are mixed; the insoluble salt precipitates immediately. General procedure: (1) Dissolve each reactant separately in distilled water. (2) Mix the two solutions — precipitate forms instantly. (3) Filter to collect the precipitate. (4) Wash the residue with distilled water to remove soluble impurities. (5) Dry the product. Ionic equation example: $Pb^{2+}(aq) + 2I^-(aq) \rightarrow PbI_2(s)$. The solubility rules card from Lesson 2 is essential for predicting which product precipitates. This method cannot be used for soluble salts because no precipitate forms.	PEDAGOGICAL NOTE — Emphasise the washing step: students often omit it. Ask: 'What soluble impurity would remain if we did not wash $BaSO_4$?' (Answer: NaCl from the spectator ions.) Use wait time of 30 seconds before cold-calling. Have students use their Lesson 2 solubility card to predict products before each demonstration — this reinforces the card as a practical tool, not just a memory exercise. Decision flowchart update: 'Is the desired salt insoluble? → Yes → Double decomposition. No → Use Lessons 3–4 methods.' Kenyan context: $BaSO_4$ is used medically as a barium meal in X-rays at Kenyatta National Hospital — safe because it is insoluble and not absorbed. PbI_2's bright yellow colour can be a memorable visual anchor. Anchor to phenomenon: 'Could CAN fertiliser be made by precipitation? Why not?' (CAN is soluble — precipitation would not work.) Students are now equipped with all three preparation methods; Lesson 6 shifts to behaviour in air.

Lesson 6 **Behaviour of Salts in Air: Hygroscopic, Deliquescent and Efflorescent Salts**	Students observe a teacher-prepared demonstration (set up at the start of the lesson or the previous day): pre-weighed samples of $CaCl_2$, $Na_2CO_3 \cdot 10H_2O$, and NaCl have been left exposed to laboratory air. Students record the mass change, visual change, and physical state change for each. $CaCl_2$: mass increased, surface appears wet, eventually a puddle forms (deliquescent). $Na_2CO_3 \cdot 10H_2O$: mass decreased, crystals became powdery white crust, crumbled (efflorescent). NaCl: no observable change (neither hygroscopic nor deliquescent nor efflorescent). Students also test silica gel (desiccant sachet) to identify a hygroscopic substance.	Three categories of salt behaviour in air: (1) Hygroscopic salts absorb water vapour from the air but do not dissolve in it — e.g., $CuSO_4$ (anhydrous, turns blue), silica gel, anhydrous $CaCl_2$. Mass increases; solid remains solid. (2) Deliquescent salts absorb water vapour from the air and dissolve completely in the absorbed water, forming a solution — e.g., $CaCl_2$, NaOH, CAN fertiliser (calcium ammonium nitrate). Mass increases significantly; solid eventually becomes a liquid. (3) Efflorescent salts lose water of crystallisation to the air (when atmospheric humidity is lower than the vapour pressure of the hydrated salt) — e.g., $Na_2CO_3 \cdot 10H_2O$ (washing soda), $CuSO_4 \cdot 5H_2O$, $FeSO_4 \cdot 7H_2O$. Mass decreases; crystals become powdery. NaCl is none of these — it is stable in air under normal Kenyan humidity conditions.	PEDAGOGICAL NOTE — THIS IS THE CENTRAL EXPLANATORY LESSON for the driving question. Return explicitly to the phenomenon: 'The fertiliser (CAN) is deliquescent — it absorbed water vapour from the night air and dissolved. The washing soda is efflorescent — it lost water of crystallisation to the dry morning air. The table salt is stable — it neither gains nor loses water under normal conditions. You can now answer the driving question.' Write this on the board and have students write it in their own words as a formative check. Common misconception: efflorescence is not the same as dissolving; the water lost was already part of the crystal structure. Differentiate hygroscopic vs. deliquescent clearly — all deliquescent substances are hygroscopic, but not all hygroscopic substances are deliquescent. Cold-call 5 students: give them a substance, ask them to classify and explain. Kenyan weather context: Nairobi's dry season vs. the long rains — humidity changes affect CAN storage; this is economically important for Kenyan farmers.
Lesson 7 **Applications of Salts in Daily Life in Kenya**	Students examine a set of real or pictured Kenyan salt-use scenarios and complete a structured case-study worksheet: (a) CAN fertiliser bags stored in a Nakuru farm store (deliquescent — must be sealed); (b) NaCl used to preserve nyama choma and fish at Lake Victoria (normal salt, stable, osmotic preservation); (c) Na_2CO_3 (washing soda) used in soap-making in Kisumu households (efflorescent — store sealed); (d) $NaHCO_3$ (baking soda) in mandazi and chapati preparation (acid salt, reacts with acid to release CO_2 for leavening); (e) $CaSO_4$ (plaster of Paris) used in Nairobi construction; (f) NaF added to Kenyan water treatment and toothpaste (dental fluorosis prevention).	The appropriate use and storage of a salt depends on its type, solubility, and behaviour in air: (1) Deliquescent salts (CAN) must be stored in sealed, moisture-proof bags; if clumped, they have absorbed humidity and may have lost nutrient concentration — they should still be usable but handled promptly. (2) Efflorescent salts (washing soda) lose water of crystallisation; while the anhydrous form is still chemically active, the change in mass means measured quantities by mass are more reliable than by volume. (3) Stable salts (NaCl) are ideal for open-air or long-term storage. (4) Acid salts ($NaHCO_3$) react with acids — useful in cooking and fire extinguishers. (5) Insoluble salts ($BaSO_4$) are safe for medical use precisely because they do not dissolve and are not absorbed by the body. Safe handling: some salts (Pb compounds, $BaCl_2$) are toxic; always wash	PEDAGOGICAL NOTE — This lesson is explicitly application and synthesis-focused. Resist the urge to introduce new chemistry; instead, reinforce transfer of concepts from Lessons 1–6. The case-study format works well in groups of 3–4; each group presents one scenario. Kenyan economic relevance: CAN is the most widely used fertiliser in Kenya; poor storage leads to crop yield losses — this is a real agricultural problem students may have witnessed at home. Prompt higher-order thinking: 'If a farmer in Eldoret opens a bag of CAN and does not use it all, what should they do and why?' Expected answer links deliquescence, sealing, and nutrient preservation. Gender and values integration: note women's roles in food preservation (NaCl for fish drying at Lake Victoria) and soap-making cooperatives. Anchor to phenomenon: 'Use everything you now know to write a

		hands; avoid inhalation of fine powders.	complete explanation of the overnight phenomenon for a fellow student who missed this unit.'
Lesson 8 **Unit Synthesis, Model Revision and Summative Assessment**	Students review their Driving Question Board (DQB) compiled across all 7 lessons. Teacher facilitates a whole-class discussion: 'What questions did we start with? Which are now answered? Are there any remaining?' Students then individually revise their cumulative particle/conceptual model of salts — originally drawn in Lesson 1 — annotating changes in understanding. The revised model must include: definition and classification of salts; solubility rules; preparation method decision flowchart; behaviour in air (hygroscopic/deliquescent/efflorescent); and one real Kenyan application per category.	Consolidated understanding across the sub-strand: (1) Salts are ionic compounds formed by full or partial replacement of hydrogen in an acid by a positive ion; classified as normal, acid, basic, or double salts. (2) Solubility rules determine which preparation method is appropriate — a non-negotiable decision step. (3) Soluble salts are prepared by acid + metal, acid + metal oxide, acid + base (titration), or acid + carbonate/hydrogen carbonate, all using excess solid (except titration) to ensure purity. (4) Insoluble salts are prepared by double decomposition (precipitation) of two soluble solutions. (5) Salts behave differently in air: deliquescent (absorb and dissolve), efflorescent (lose water of crystallisation), hygroscopic (absorb without dissolving), or stable. (6) These properties directly determine how Kenyan farmers, cooks, builders, and health workers should store and use salts safely and effectively.	PEDAGOGICAL NOTE — This lesson is assessment and metacognition-focused; do not introduce new content. The model revision is a key formative-to-summative bridge: collect models as assessment evidence. Summative assessment should include: (a) a definition/classification question; (b) a solubility-rules application question; (c) a preparation method selection and equation question; (d) a behaviour-in-air classification question with explanation; (e) a Kenyan-context application question (e.g., advise a Rift Valley maize farmer on CAN storage). Return explicitly and visibly to the driving question — write it on the board, have 2–3 students give a full oral answer, then have all students write their final answer independently. Celebrate the intellectual journey: 'On Day 1 you could not explain why the fertiliser clumped. Today you can explain it at the particle level and give practical advice to a Kenyan farmer.' This closes the anchoring phenomenon loop and reinforces the value of scientific understanding.